

Adroit Technologies

Protocol Driver

Name	OPC UA Client
DLL	D_OPC_UA



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Contents

1. Overview	4
1.1. Connectivity	4
1.2. Driver Installation and Setup.....	4
1.3. OPC UA Instance Certificates	4
1.4. OPC UA Server Certificates	5
2. Wiring	6
3. Device Configuration	7
3.1. Device Configuration	7
3.1.1. OPC UA Server	7
3.1.2. General.....	10
3.1.3. Test Connection.....	11
3.2. Connect To Server.....	11
3.2.1. Server Discovery Service	11
3.2.2. Connections Details.....	12
4. Registry Details.....	13
5. Protocol Definition and Considerations	14
5.1. Scanning Configuration.....	14
5.2. Message Protocol	14
6. General Notes and Troubleshooting	15
6.1. Contact Details.....	15
6.2. Adroit Technologies Knowledge Base.....	15
6.3. Troubleshooting	15
6.3.1. Datascope Errors	15
6.3.2. Message Boxes.....	16
7. Revisions	17
7.1. Document Revision.....	17
7.2. DLL Revision	17

1. Overview

OPC UA (OLE for Process Control Unified Architecture) is a standard specification that makes it possible for automation/control software applications and field systems/devices to interoperate and communicate data between each other.

To learn more about what OPC UA is, visit the OPC foundation website:

<https://opcfoundation.org/about/opc-technologies/opc-ua/>

The OPC UA client driver for Adroit communicates to OPC servers that expose an OPC UA interface.

1.1. Connectivity

The OPC UA client driver communicates with OPC UA Servers in accordance with the OPC UA Specification whose details are beyond the intended scope of this document.

1.2. Driver Installation and Setup

If the Adroit OPC UA Client driver is not installed, you will need to install it. Installation of this driver for the first time will install the OPC Foundation's OPC UA redistributable if not already installed on your PC.

The following files will be installed agent server folder:

- d_opc_ua.dll
- d_opc_ua.doc
- opcuabrowser.exe

as well as the following 3rd party dependency files:

- libeay32.dll
- libxml2.dll
- TB5CPP.dll
- TB5OT.dll
- TB5STACK.dll
- TB5UTIL.dll

If any of these files is not in the agent directory, then the driver did not install properly.

1.3. OPC UA Instance Certificates

The OPC UA specification requires that all UA clients have an instance certificate that positively identifies the client to the server. The instance certificate for Adroit OPC UA is automatically created on the Agent Server PC when the OPC UA Protocol driver DLL is loaded by PSEXEC or AS.EXE for the first time.

The instance certificate can also be created manually by executing the *%ProgramData%\Adroit Technologies\Adroit\certs\create_instance_certificate.bat* file which is found in the folder. This will be required if the certificate expires. Before executing the batch file, delete the existing certificate called *cert_adroit_client.der* file will be found in the *%ProgramData%\Adroit Technologies\Adroit\certs\own* folder.

Once the batch file has completed, the following folders will have been created under the *certs* folder, if the batch file is being executed for the first time:

crl
issuer
own
private
rejected
trusted

As mentioned above, the *cert_adroit_client.der* file will be found in the *certs/own* folder and the *private_key_adroit_client.pem* file will be found in the *certs/private* folder. Select these two files for the instance certificate entries above.

Furthermore, the *cert_adroit_client.der* file needs to be copied to the trusted certificate folder of the OPC UA Server/s that you will be communicating to. Conversely, the OPC UA Server instance certificate needs to be copied to the Adroit system folder for two way authentication. The certificate types should be the same.

1.4. OPC UA Server Certificates

The OPC UA specification requires that all UA clients trust the server certificate for the OPC UA server that they are connecting to.

By default, server certificates will be placed in the *%ProgramData%\Adroit Technologies\Adroit\rejected* folder when a client connects to the OPC UA server for the first time.

These must be manually copied to the *%ProgramData%\Adroit Technologies\Adroit\trusted* folder before a successful connection will be allowed.

2. Wiring

Not applicable.

3. Device Configuration

When configuring an OPC UA client device, the following dialog is launched and must be completed accordingly.

3.1. Device Configuration

3.1.1. OPC UA Server

On selecting the [...] button for "Endpoint URL" you will be presented with the "Connect to Server" dialog. See section [Connect to Server](#).

The device configuration dialog has 3 different layouts depending on the authentication type selected.

Anonymous authentication (Default):

OPC UA Client Protocol Driver - OPCUA

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
OPC UA Server	
Endpoint URL	
Name	
Authentication (token) type	ANONYMOUS_0
Message mode	None
Policy	http://opcfoundation.org/UA/SecurityPolicy#None
General	
Timestamp source	Local
Polled	False
Timeout (ms)	0
Retries	0
Maximum number of groups	0
Maximum no of tags per group	0
Read after write	False
Use default sample rate	False
Test connection	
Test	

Authentication (token) type
The type of security token to be used

Close

Username Authentication looks as follows:

OPC UA Client Protocol Driver - OPCUA

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
OPC UA Server	
Endpoint URL	
Name	
Authentication (token) type	USERNAME_1
Message mode	None
Policy	http://opcfoundation.org/UA/SecurityPolicy#...
User credentials (if authentication type = UserName)	
Username	
Password	
General	
Timestamp source	Local
Polled	False
Timeout (ms)	0
Retries	0
Maximum number of groups	0
Maximum no of tags per group	0
Read after write	False
Use default sample rate	False
Test connection	
Test	

Authentication (token) type
The type of security token to be used

Close

User credentials - if authentication type = *username*, a username and password must be specified.

Certificate Authentication looks as follows:

OPC UA Client Protocol Driver - OPCUA

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

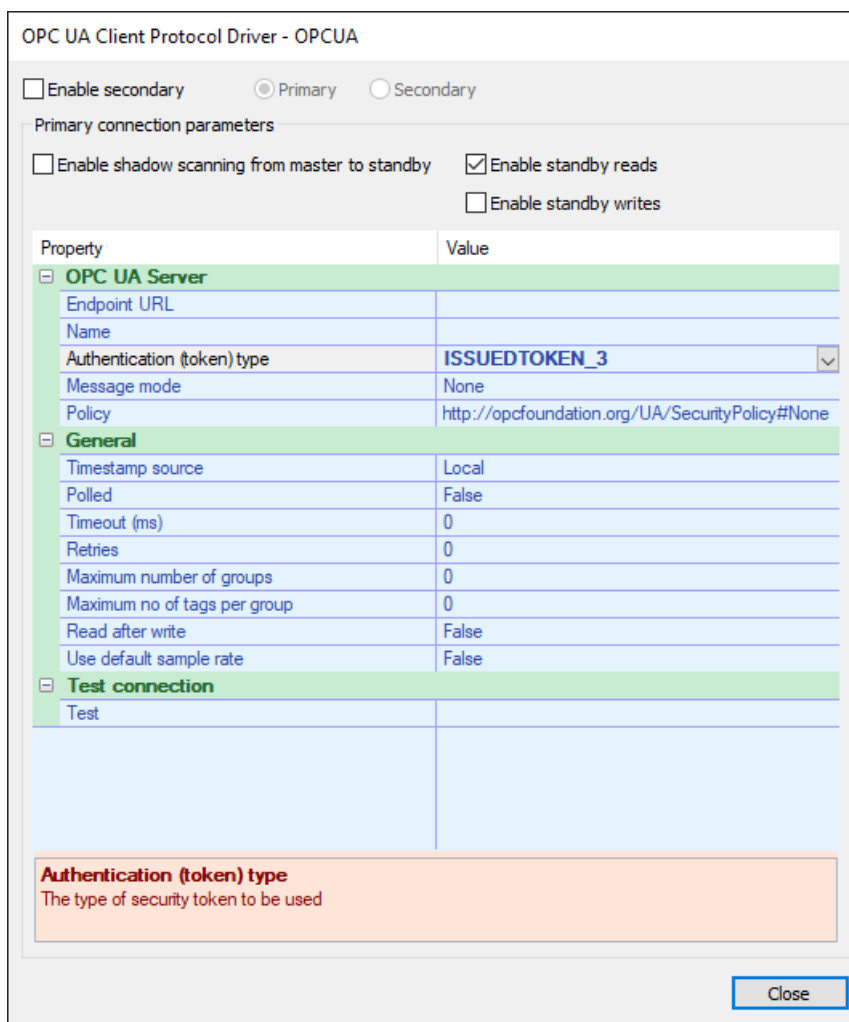
Property	Value
OPC UA Server	
Endpoint URL	
Name	
Authentication (token) type	CERTIFICATE_2
Message mode	None
Policy	http://opcfoundation.org/UA/SecurityPolicy#...
User certificate (if authentication type = Certificate)	
Certificate file	
Private key file	
Private key password	
General	
Timestamp source	Local
Polled	False
Timeout (ms)	0
Retries	0
Maximum number of groups	0
Maximum no of tags per group	0
Read after write	False
Use default sample rate	False
Test connection	
Test	

Authentication (token) type
The type of security token to be used

Close

User certificates - if authentication type = *certificate*, a certificate must be specified. User certificates need to be issued by a 3rd party certification authority.

Issued Token Authentication looks as follows:



OPC UA Client Protocol Driver - OPCUA

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
OPC UA Server	
Endpoint URL	
Name	
Authentication (token) type	ISSUEDTOKEN_3
Message mode	None
Policy	http://opcfoundation.org/UA/SecurityPolicy#None
General	
Timestamp source	Local
Polled	False
Timeout (ms)	0
Retries	0
Maximum number of groups	0
Maximum no of tags per group	0
Read after write	False
Use default sample rate	False
Test connection	
Test	

Authentication (token) type
The type of security token to be used

Close

3.1.2. General

Timestamp source – Local, Server or Source. Default is Local.

- Local – If this option is selected, the driver will update values with the timestamp of when the driver receives the value.
- Server – When the OPC server sends a value the driver, whether by exception or from being polled, the OPC server return two timestamps, the timestamp that the server received the value, or the timestamp that the device/source received the value, by selecting this radio button, the driver will update values with the server's timestamp.
- Source – The driver will update values with the device/source timestamp as mentioned above. (**Important:** do not use this timestamp option if the communication protocol connecting to the device does not supply a timestamp)

Should you wish to use Server or Source timestamps, you must ensure that the agent server and the OPC UA Server and/or the remote device are time synced to the same time source as the agent server!

Polled – OPC clients by nature receive data "by exception" from OPC Servers. However, you can use OPC for polling data from front end devices. This setting enables polling as opposed to "by exception". This setting should only be used if absolutely necessary. (Default off)

Timeout – The timeout value in msecs. If polled mode is not enabled (which is the norm), you must either:

- Configure a watchdog tag whose scan rate is smaller than the timeout value or
- Disable the timeout value i.e., 0

Retries – The number of retries in polled mode before switching to connections between primary and secondary connections.

Maximum number of groups – The maximum number of groups to be created. A value of 0 means no restriction.

Maximum no. of tags per group – The maximum number of tags allowed per group. A value of 0 means no restriction.

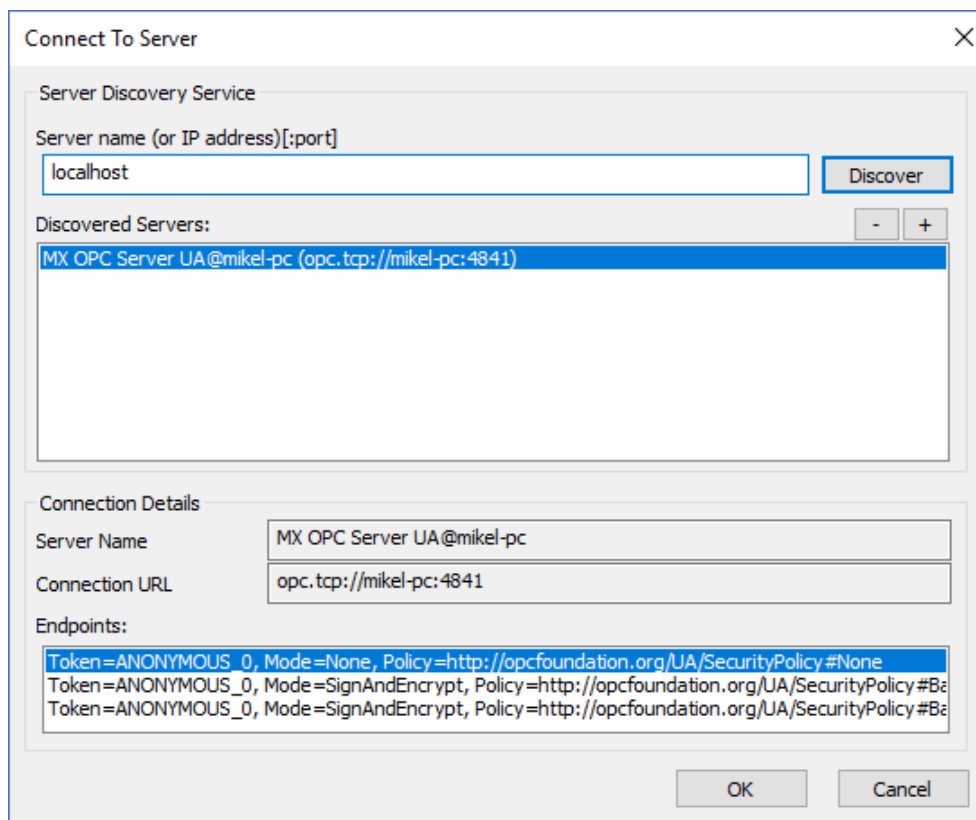
Read after write – Always perform a confirmation read tag from OPC Server after writing a new value.

Use default sample rate – Uses the sample rate of the item on the OPC Server as the scan rate, rather than the configured scan rate setup in Adroit Configurator.

3.1.3. Test Connection

Test – Test the connection to the OPC UA server using the configured endpoints. If the connection to the OPC UA server is being made for the first time, you will be prompted as to whether you wish to trust the server certificate.

3.2. Connect To Server



3.2.1. Server Discovery Service

Server name (or IP address) [:port] – This is the name or IP address of the machine on which an OPC UA server discovery will be conducted. To initiate a discovery on the machine or IP address entered, click on the refresh button.

Discovered Servers – This list displays all available OPC UA servers which are discoverable on the machine or IP address specified in “Discovery Host Machine.”

The [+] button allows you to enter a server manually that is not discoverable on the network if the format *opc.tcp://hostname:port* e.g. *opc.tcp://klaus:1234*.

Manual entries are stored as a CSV entry in the registry value:

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\OPC UA\Manual.

The [-] button allows you to remove the currently selected server if it is a manually added server.

If any of the servers displayed in “Discovered OPC Servers” are clicked on, then the “Connection Details” will automatically populate with the selected server’s details.

3.2.2. Connections Details

Server Name – The name of the server to which the current device will connect to.

Connection URL – The server endpoint address on which the device will connect when connecting to the OPC UA server.

Endpoints – The list of endpoints (authentication methods) supported by the selected OPC UA Server. Contains the definitions for the token type, message security mode and security policy.

The combination of token type, message security mode and security policy must be valid for the OK button to be enabled.



PLEASE NOTE

It is best to always make your selection and press [OK] on this “Connect to Server” dialog because pressing [Cancel] will make all the “OPC UA Server” group of properties editable. Editing these properties manually is done at your own risk.

4. Registry Details

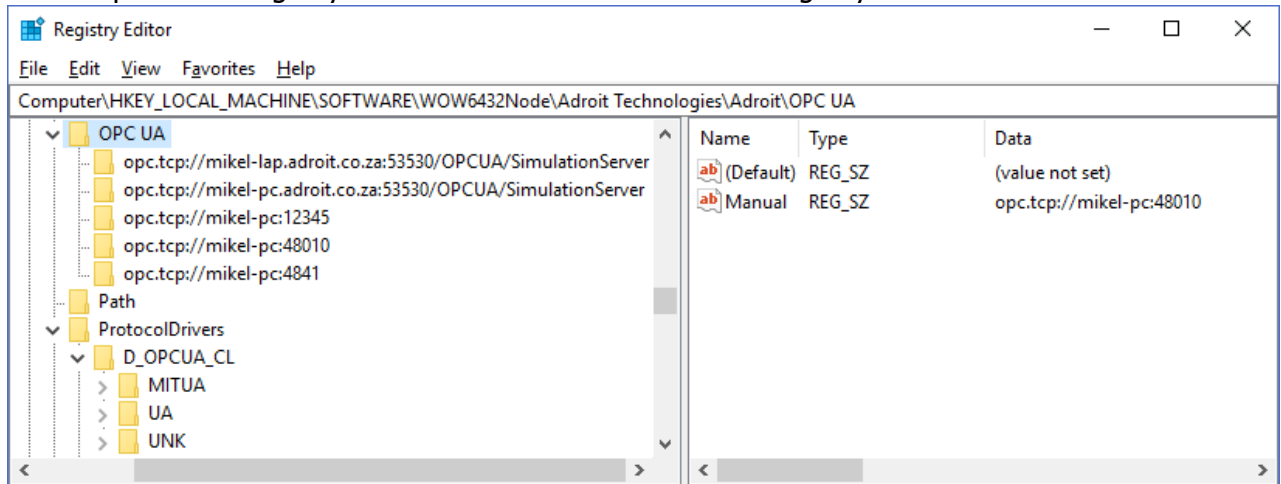
As is the case with all Adroit Protocol Drivers the configuration for a device is stored in the registry at:

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\D OPCUA_CL\DeviceName

However, to allow both the driver and opcuabrowser.exe to share connection details for OPC UA Servers, these details are stored at :

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\OPC UA\ConnectionURL

An example of the registry store is show below where both registry sections are visible:



5. Protocol Definition and Considerations

5.1. Scanning Configuration

The default address is *ns=1;s=address*.

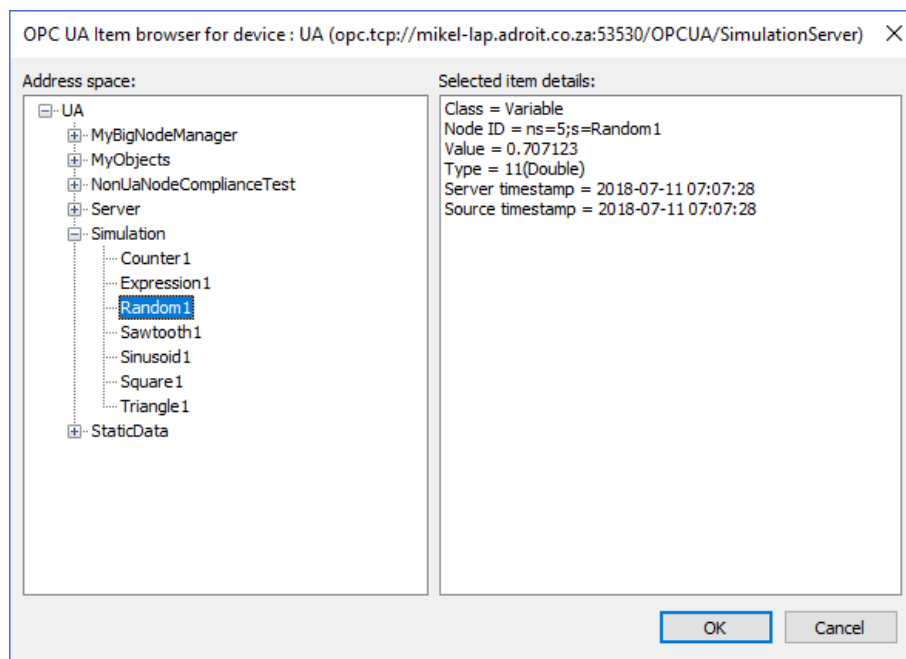
Each node in the OPC UA address tree is made up of an address tuple which contains the namespace index and address of the node.

The OPC UA address format is as follows:

ns=x;s=y

Where x is the namespace index of the node and y is the address of the node respectively.

A node's namespace index and address can found/browsed by using the OPC UA Browser. Clicking the browse button in the scanning dialog will launch an item browser (not opcuabrowser.exe!) which will browse the entire address tree on the connected OPC UA server. The browser dialog looks as follows:



- 1.
2. If the driver is in polled mode, the update rate of a tag is determined by the scan period specified. Please note :
 - This driver does not support dual scanning. In other words should you scan more than one tag to a single address, subsequent tags will always remain scan bad.
 - This driver does not support "Output Initialize".

5.2. Message Protocol

Datascopes messages are displayed in ASCII mode.

The underlying communication protocols to the device is not available to the OPC Client and thus not showed, see the [troubleshooting](#) section.

This window will however show all validation of tags. If the tag cannot be validated, a message indicating the error message will be displayed here.

All item updates (process-data changes) are also indicated here with an offset number of the tag that's being updated.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

6.3.1. Datascope Errors

Cannot connect to server	Ensure that the OPC UA Server is running
Re-connect failed (ERROR : Failed to get endpoints: 0x80AD0000(BadDisconnect))	Ensure that the OPC UA Server is running
Re-connect failed (ERROR : Failed to connect the session: 0x801A0000(BadCertificateUntrusted))	Ensure that the certificate of the OPC UA Server is trusted by Adroit by moving the certificate from the rejected folder to the trusted folder
Re-connect failed (ERROR : Failed to connect the session: 0x80130000(BadSecurityChecksFailed))	Ensure that the Adroit certificate is trusted by OPC UA Server
Re-connect failed (ERROR : Failed to get endpoints: 0x80AD0000(BadDisconnect))	Ensure that the OPC UA Server is running Ensure that the certificate of the remote OPC UA Server is trusted

6.3.2. Message Boxes

ERROR : Failed to find servers at opc.tcp://hostname:4840	1. The host does not exist OR 2. The host is not locatable by your DNS Server - try using the IP Address instead of the <i>hostname</i>. 3. The OPC UA Local Discovery Service is not running on the host. 4. If <i>hostname</i> is the local PC - there is a known issue caused by Windows' preference of IPV6 over IPV4. See https://theitbros.com/ping-returns-ipv6-address-ping-ipv4/ for more information on the subject which will tell you how to set IPV4 over IPV6. Otherwise, simply use the IP address of the host instead of the host name.
The OPC UA Server certificate is not trusted by this server. Would you like to trust this OPC UA Server?	If you press [Yes] Adroit will trust the OPC UA Server for you.
Our client certificate is not trusted by the OPC UA Server. Please invoke the trust on the OPC UA Server (or move the rejected client certificate to the trusted folder on the OPC UA Server	If you press [Yes] Adroit will trust the OPC UA Server for you.
ERROR: Failed to get endpoints: 0x803D0000 (BadNotSupported)	

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	16-Jan-2022	K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
	27-May-2015	First Draft
	03-Jul-2018	Added OPC UA browser section
	16-Oct-2018	Removed OPC UA browser section
51	14-Oct-2022	Do not poll if not Reads Enabled (clustering) Set scan bad in polled mode if quality is not good Ignore OPC Server value callbacks if not Reads Enabled Added Load configuration (dynamic config) and at start items will mark healthy. Force readAge to 0 in ReadItem() so plc is read and not cache Implemented SetLastTransactedValue() and IsPrevValueStale() update checks Added PrepareForFirstPoll() if in polled mode, to reinitilised poll timers, ResetConnection included Remove that items didnt clean up groups in polled mode during runtime

